

Business Problem Statement

- **Optimizing Supply Chain Efficiency: Reducing Perishable Waste and Protecting Margins**

Context & Background

In the retail supply chain of perishable goods, balancing product availability with expiration risks is a critical operational challenge. This dataset tracks sales, profits, and inventory metrics across various perishable categories—including high-value, strict-expiry items like Pharmaceuticals and high-turnover consumer goods like Frozen Foods—across multiple regional store locations.

The Core Business Problem

The organization is facing a severe profitability drain driven by an unsustainable inventory leakage rate. While the business generates a strong top-line gross revenue of ₹59.11 Crores, the Average Profit Margin is heavily compressed at just ₹13.39.

The primary driver of this low margin is an aggressive inventory mismatch: Total Unit Waste has reached 5,201,615 units against 20,283,565 Total Units Sold. This indicates that roughly 20.4% of all product stock is lost to waste/spoilage, representing massive unrecovered capital, high holding costs, and logistics inefficiencies.

Key Operational Complications

1. Severe Revenue Asymmetry: Over 68% of total revenue (₹40.78 Crore) is generated solely by the Pharmaceuticals category, creating a risky single-point-of-failure dependency.
2. Performance Stagnation Across Stores: While STORE_040 has optimized its local supply chain to achieve peak profits (₹27.40 Lakhs), all other stores (STORE_012, STORE_017, STORE_032, STORE_028) are underperforming and stagnant, clustering uniformly around lower profit thresholds (~₹22 Lakhs).
3. Product-Specific Waste Leaks: Volume-heavy categories like Frozen Dinners and Frozen Desserts are suffering from the highest unit spoilage rates, indicating a failure in local demand forecasting or cold-chain management.

Project Objectives

- To aggregate fragmented sales and spoilage data into a centralized data model.
- To identify high-waste operational bottlenecks across product lines and store locations.
- To deliver data-driven recommendations that will reduce the 20.4% wastage rate, optimize seasonal inventory during demand fluctuations (such as the October demand dip), and scale the operational blueprint of high-performing stores.